



SIMU-VERSE

Interactive Physics Simulation Platform

Learn physics through AI-guided lessons, real-time simulations, and interactive visualisations.

🌟 Overview

SIMU-VERSE is a web-based educational platform that makes physics tangible. It combines **AI-generated lessons**, **interactive simulations**, **YouTube video support**, and a **book-style reference slideshow** – all wrapped in a futuristic, responsive UI.

Live Demo: <https://simuverse-team.netlify.app/>

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🎯 Features

Feature	Description
🔑 Authentication	Sign up / Log in using Supabase (email/password).
📖 AI Lesson Slides	Dynamic slide decks generated via Anthropic Claude (fallback to built-in content).
🎮 Interactive Simulations	Real-time canvas-based simulations for Wave Optics, Projectile Motion, and Electrostatics.
🗣️ AI Chatbot	Context-aware physics tutor (NOVA) using Gemini API with conversation memory and throttling.
📺 YouTube Video Block	Inline YouTube player – no redirects, fully themed.
📖 Book References Modal	Separate slideshow for high-resolution PDF-extracted images (per topic).
🎨 Dual Theme	Login screen retains original neon colours; main app uses a softer, high-contrast theme.
📱 Responsive	Works on desktop, tablet, and mobile.

Physics Topics Covered

1. Wave Interference & Diffraction

- Young’s Double Slit Experiment
- Single Slit Diffraction
- Path difference, fringe width, intensity distribution
- Interactive wave simulation with adjustable λ , d , D

2. Projectile Motion & Energy Conservation

- Oblique projection (2D motion)
- Kinematic equations & trajectory parabola
- Real-time energy graph (KE / PE / TE)
- Adjustable velocity, angle, mass, gravity

3. ⚡ Electric Field & Potential

- Single and multiple point charges
- Electric field lines & equipotential surfaces
- Coulomb’s law and superposition
- Drag-and-drop charges, test charge force vector

Tech Stack

Area	Technologies
Frontend	HTML5, CSS3, Vanilla JavaScript
Authentication	Supabase (Auth)
AI Lesson Generation	Anthropic Claude API (optional)
AI Chatbot	Google Gemini API (gemini-2.0-flash-lite)
Hosting	Netlify (recommended) / GitHub Pages
Icons & Fonts	Google Fonts (Inter, JetBrains Mono, Rajdhani, Orbitron)
Vector Graphics	Inline SVG (diagrams) & Canvas API (simulations)